



# Resting Electrocardiogram System

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# Outline

**1. Introduction**

**2. Background & Motivation**

**3. Components & Functions**

**4. Hardware Testing:** Notch Filter and Low Pass Filter

**5. Software Verification:** Simulink

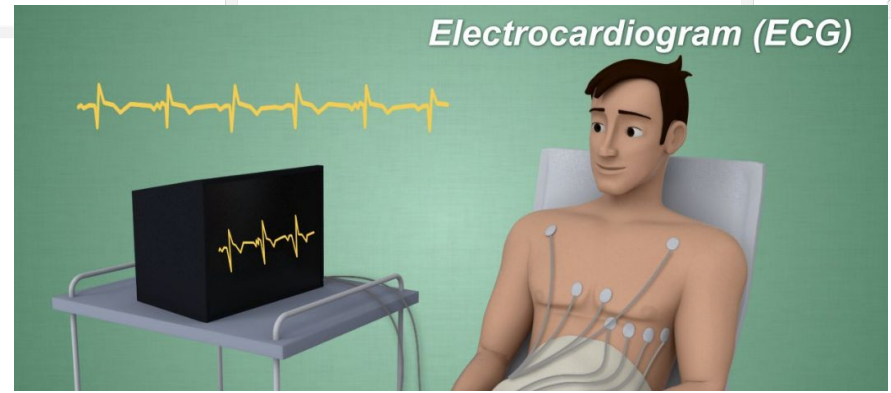
# Introduction

01.

# Introduction

**Electrocardiogram (ECG):** A test that records the electrical activity of the heart using surface electrodes.

- It reflects **cardiac rhythm, rate, and conduction patterns**, helping clinicians detect abnormal heart function.



## Signs & Symptoms Indicating the Need for ECG

- Chest pain or discomfort
- Dizziness or fainting
- Palpitations (irregular or rapid heartbeat)
- Shortness of breath
- Fatigue



**Cardiovascular diseases are the leading global cause of death, claiming ~17.9 million lives annually. Early ECG use can help reduce this toll.**

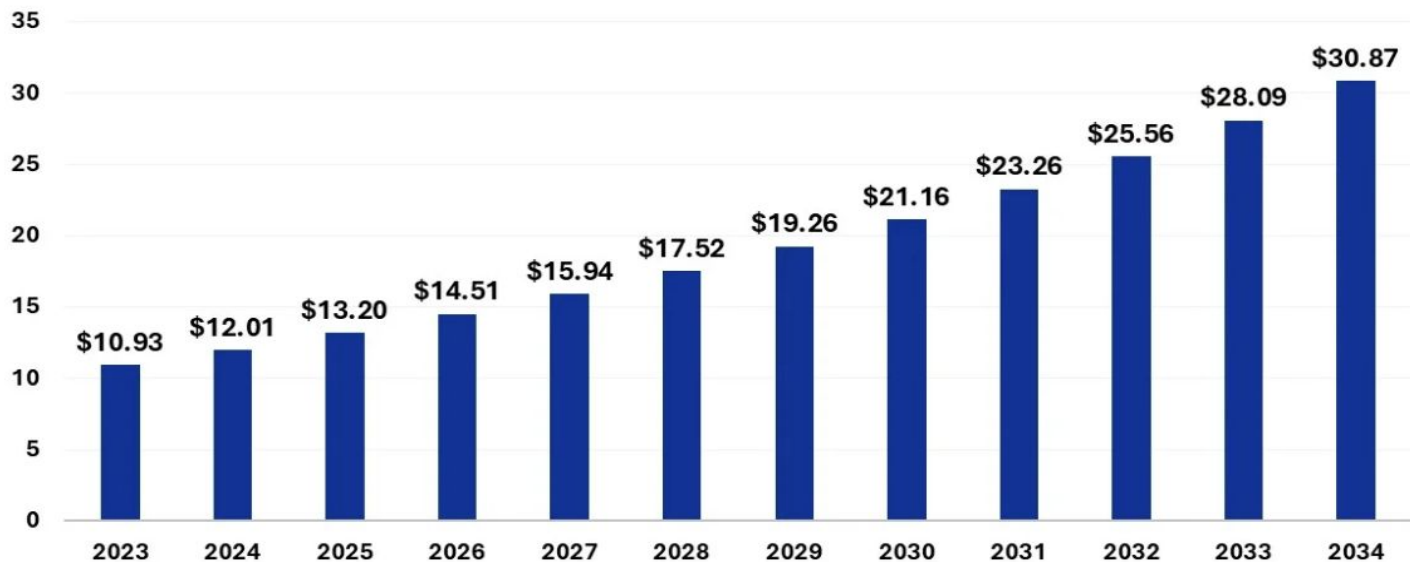
# ECG Stat Facts



In the United States:

- One person dies **every 33 seconds** from cardiovascular disease.
- In 2022, **702,880 people died from heart disease**. That's the equivalent of 1 in every 5 deaths.
- Heart disease cost about **\$252.2 billion** from 2019 to 2020. This includes the cost of healthcare services, medicines, and lost productivity due to death.

## Electrocardiograph (ECG) Market Size 2023 to 2034 (USD Billion)



Source: <https://www.precedenceresearch.com/electrocardiograph-market>

# ELECTROCARDIOGRAM TIMELINE



The term 'electrocardiogram' used to describe these wave forms was first coined by Einthoven at the Dutch Medical Meeting.

The first electrocardiogram machine was introduced to the United States in 1909 by Dr. Alfred Cohn at Mt. Sinai Hospital, New York.

The American Heart Association and the Cardiac Society of Great Britain published their recommendation for recording the exploring lead from six sites named V1 through V6 across the precordium.

Researchers from Texas show that 12-lead ECGs transmitted via wireless technology to hand-held computers is feasible and can be interpreted reliably by cardiologists.

Apple Inc. Launches apple watch for EKG reading and prevention of arrhythmias.

1891

1893

1901

1909

1930

1938

1942

1954

1976

1999

2006

2018

Einthoven, successfully developed a new string galvanometer with very high sensitivity, which he used in his electrocardiograph. A 600 pounds device

Dr. Emanuel Goldberger, constructed unipolar leads with the central terminal and connected to additional positive unipolar leads on each of the left and right arms and the left leg.

Danish cardiologists report the successful reduction in the time between onset of chest pain and primary angioplasty when the ECG of patients is transmitted wirelessly from ambulance to the cardiologist's handheld PDA (Personal Digital Assistant).

William Bayliss and Edward Starling, demonstrated triphasic cardiac electrical activity in each beat using an improved capillary electrometer.

The importance of electrocardiogram in differentiating cardiac from non-cardiac chest pain was well recognized.

Erhardt and colleagues describe the use of a right-sided precordial lead in the diagnosis of right ventricular infarction.



# Research Gaps in ECG Systems

## Signal Quality & Noise Interference

- Baseline wander and high-frequency noise reduce diagnostic accuracy.

## Lead Misplacement & User Error

- Incorrect electrode placement impacts waveform, leading to misinterpretation.

## Data Management Challenges

- Difficulty in integrating with modern **Electronic Health Record (EHR)** systems.

## Patient Compliance

- Patients may avoid regular testing due to uncomfortable electrode patches.



**Addressing these issues is critical to improve accuracy, clinical reliability, and integration in digital healthcare.**



# Motivation & Objective

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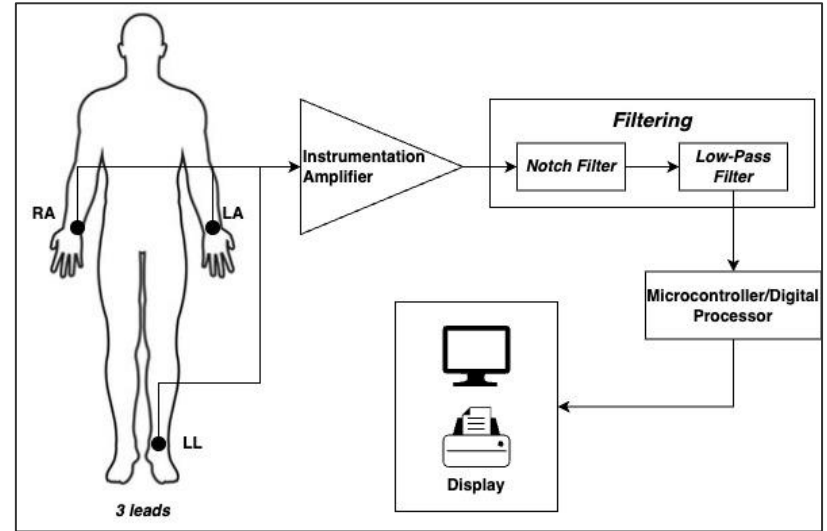
# Motivation & Objectives

**Enhance** hardware-level signal

conditioning

- 1) **Design** a high-gain amplification circuit
- 2) **Integrate** a 60 Hz notch filter
- 3) **Implement** a 150 Hz low-pass filter
- 4) **Ensure** accurate and clean ECG

acquisition



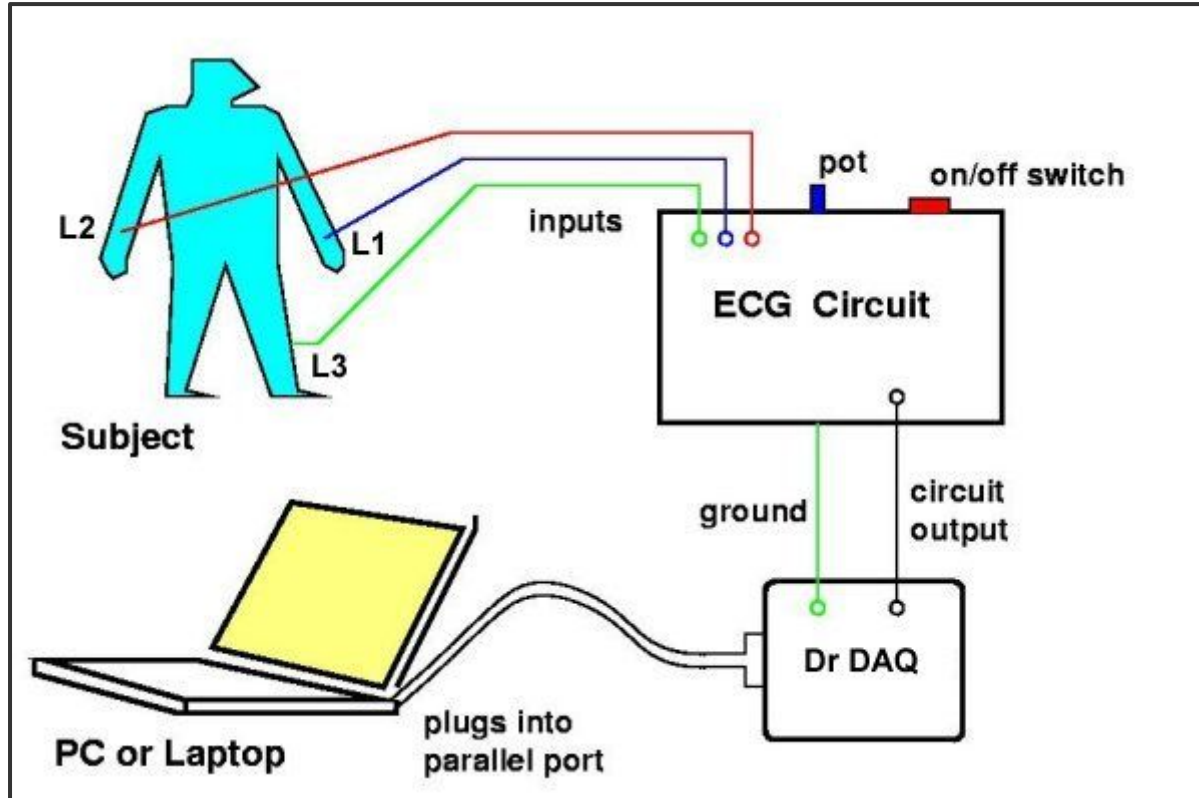
Proposed Circuit

# Components and Function

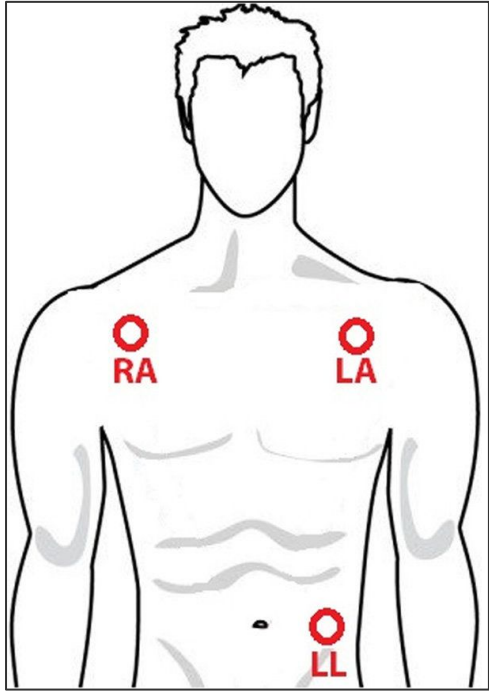
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# ECG System Components



# 1st Component: *Electrodes*



Purpose: Detect bioelectric signals from the skin

3 Leads:

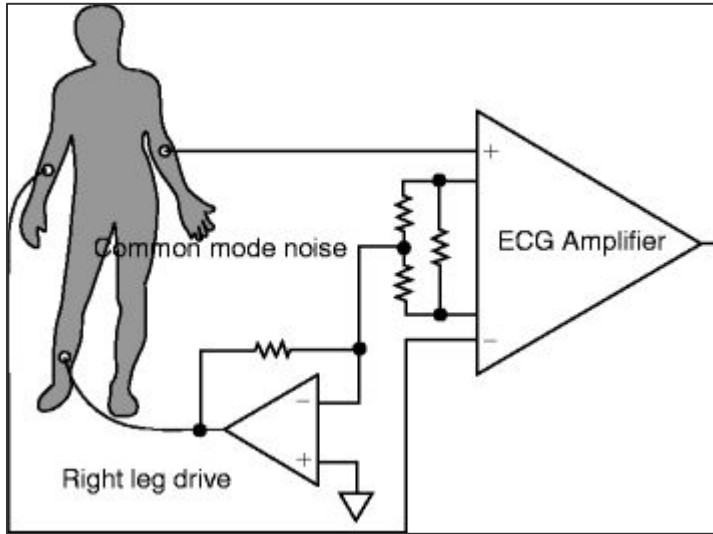
1. RA - Right Arm
2. LA - Left Arm
3. LL - Left Leg

→ *Signal Input*

→ *Ground*

Material: Ag/AgCl gel electrodes

# 2nd Component: *Instrumentation Amplifier*



Purpose: Amplifies small ECG signals with high precision

High Common Mode Rejection Ratio (CMRR):

- Ignores noise common to both input wires
- **Example:** If both RA and LA inputs pick up the same 100mV noise the CMRR rejects it

ECG Specific Design Requirements:

- Gain > 1000 is typically used to amplify physiological signals

# 3rd Component: *Notch Filter*

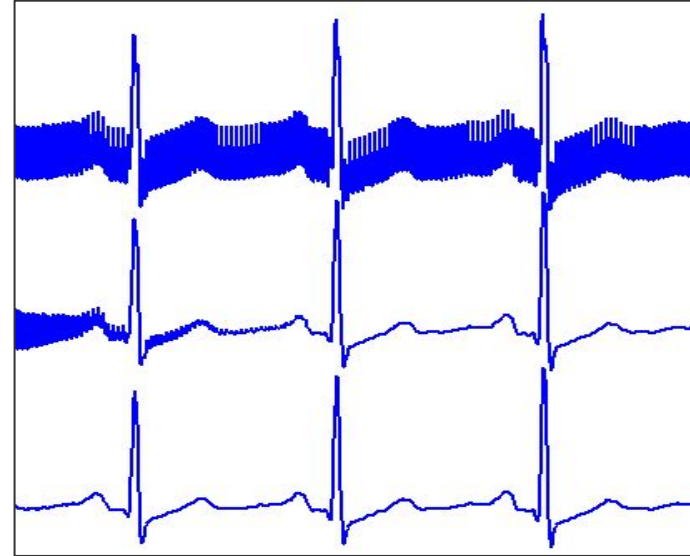
Definition: A narrow-band filter that suppresses interference from power lines

ECG Specific Design Requirements:

- Cut-Off Frequency = 60 Hz
- Bandwidth = 5 - 10 Hz

Purpose:

- Removes ambient power line interference
- Essential for ECG Waveform



<https://arturomoncadatorres.com/ecg-iir-notch-filtering-using-transient-suppression/>



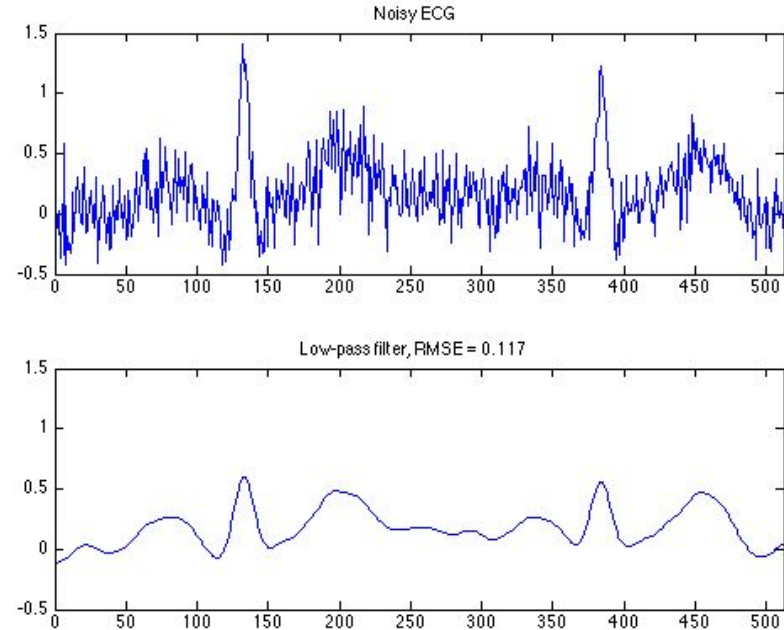
# 4th Component: *Low Pass*

## Purpose:

- Removes muscle noise and high-frequency artifacts
- Preserves key ECG features

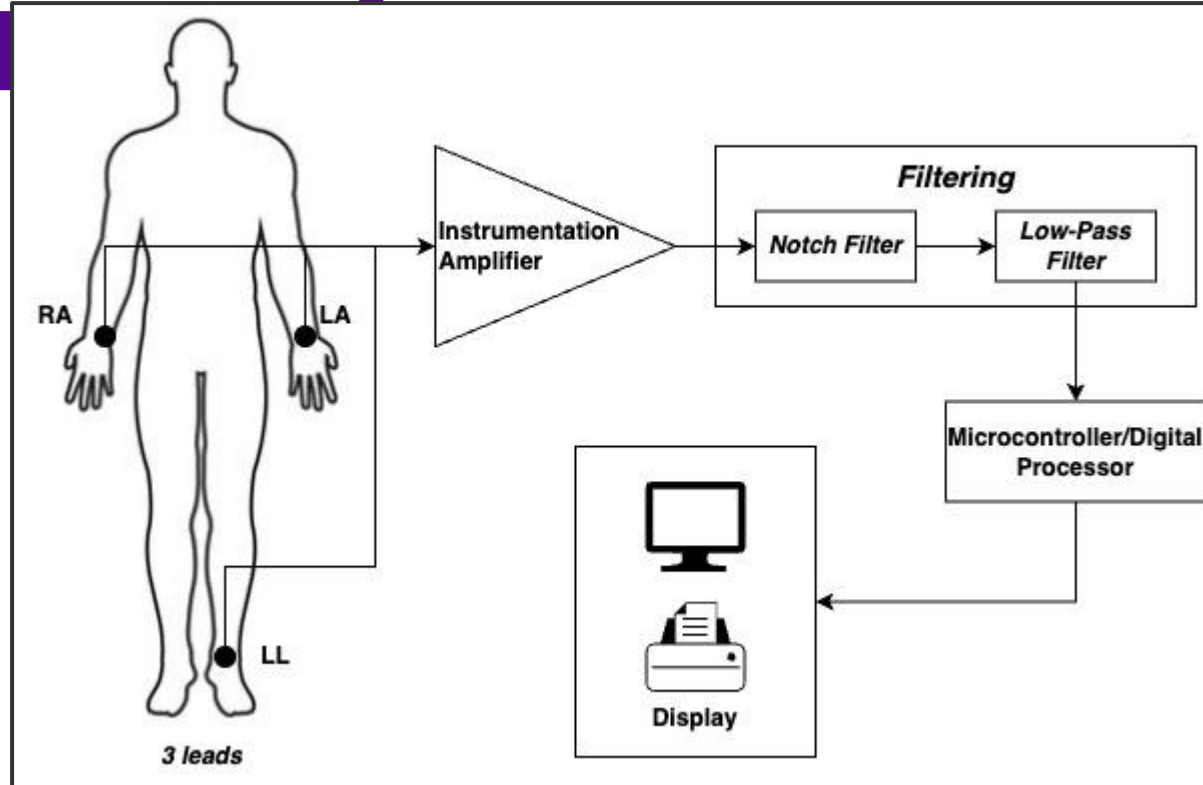
## ECG Specific Design Requirements:

- Cutoff Frequency ~150 Hz as per American Heart Association Guidelines
- Second-order low pass filter
- Gain = 1



# Our ECG System

## Comp

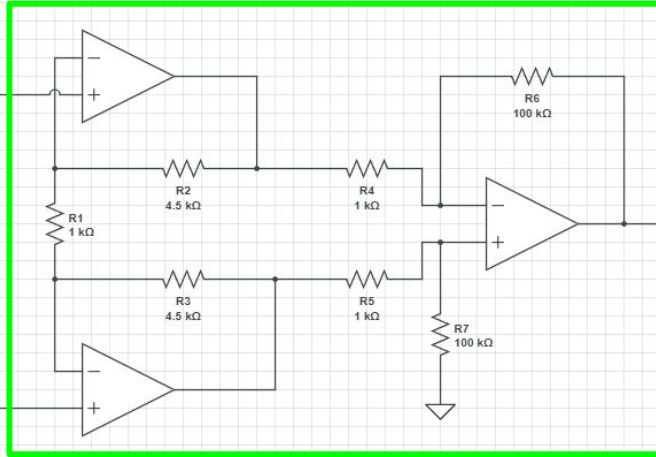


# Hardware Testing

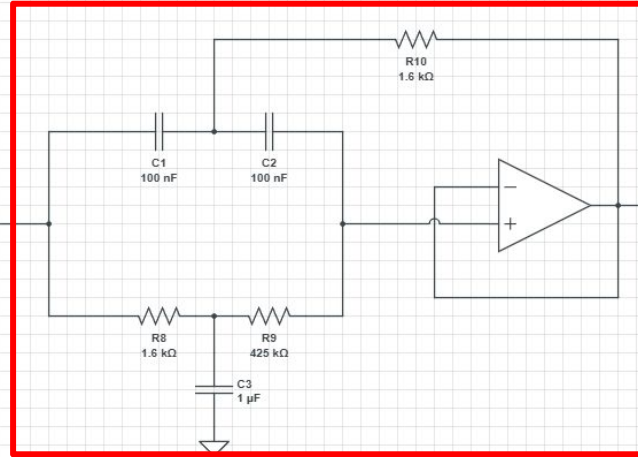
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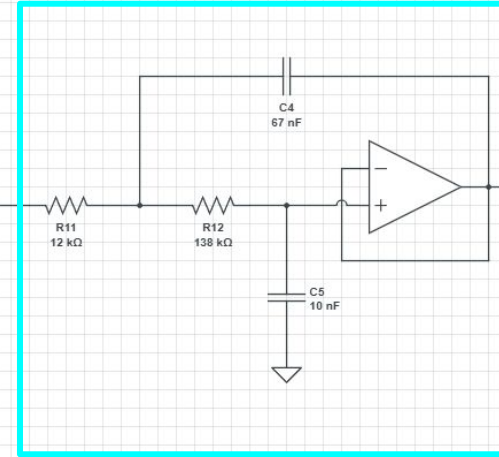
# Hardware Testing



Instrumentation  
Amplifier (INA)

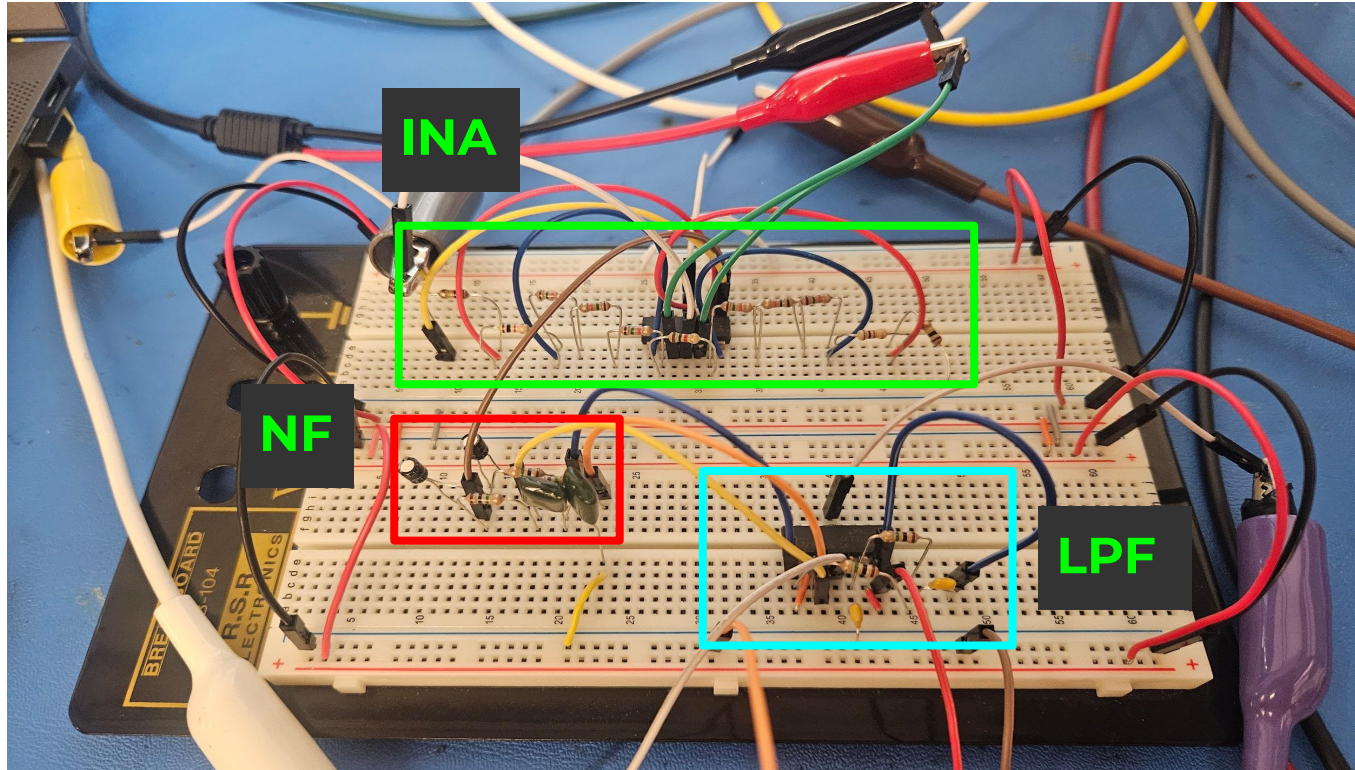


Notch Filter (NF)

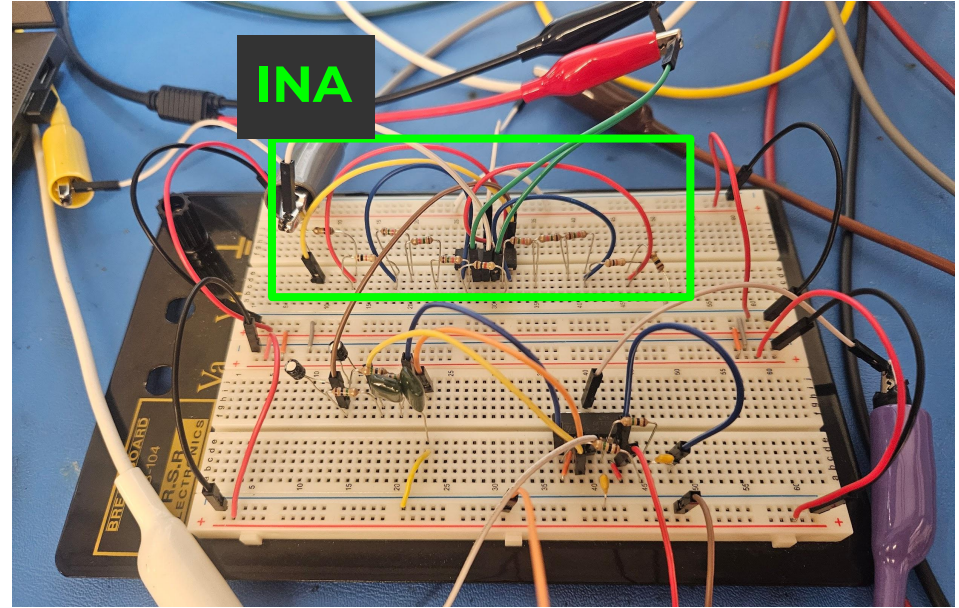
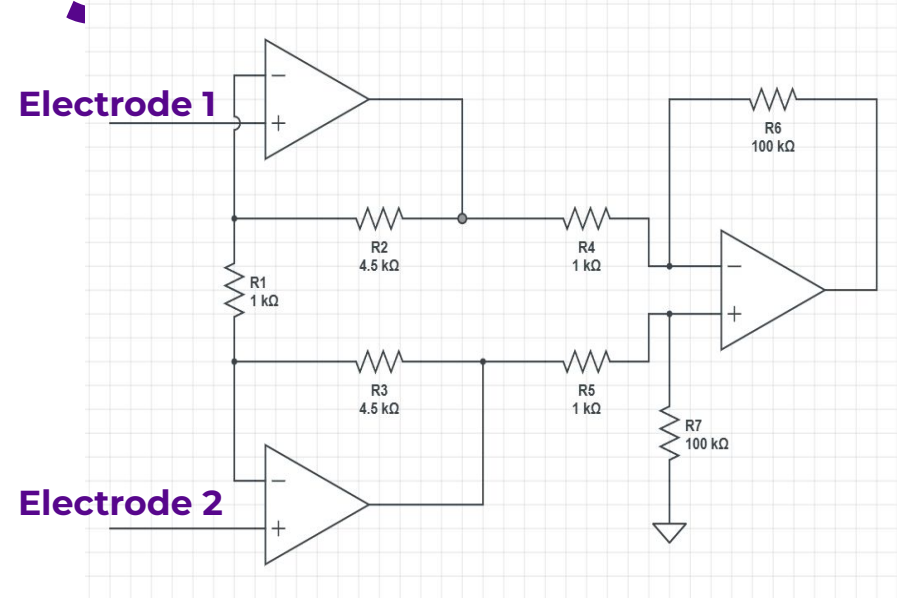


Low Pass Filter  
(LPF)

# Hardware Testing



# Instrumentation Amplifier Setup



Electrode 3  
(Ground)

$$R1 (R_{gain}) = R4 = R5 = 1K\Omega$$

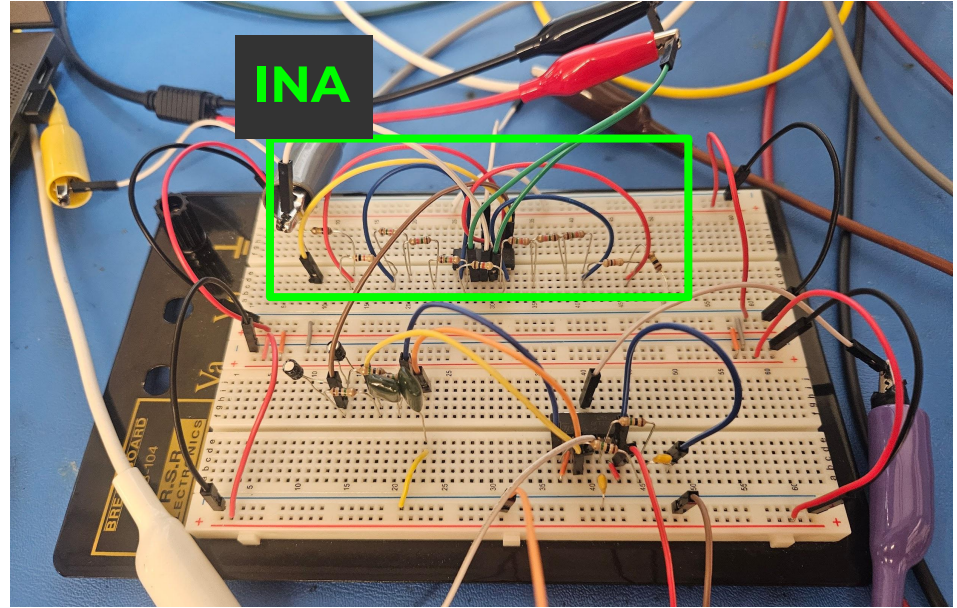
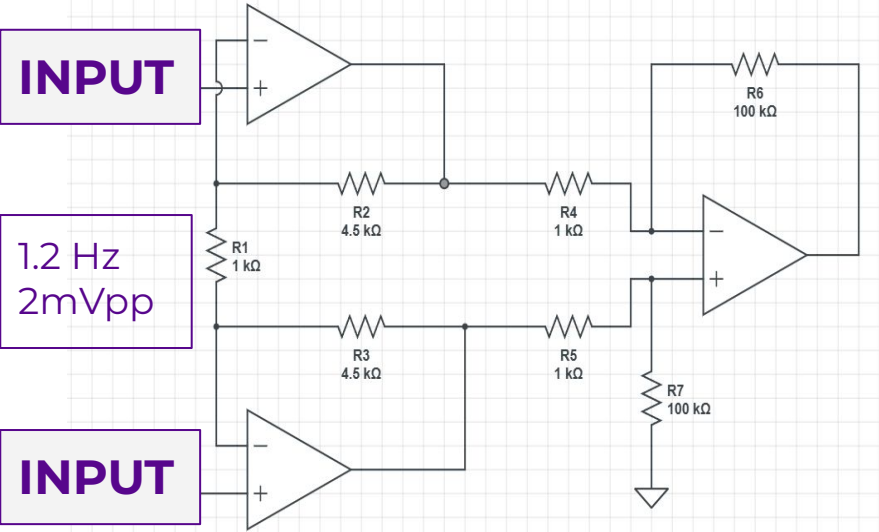
$$R2 = R3 = 4.5 K\Omega$$

$$R6 = R7 = 100K\Omega$$

$$\text{Gain} = \left(1 + \frac{2 * R_2}{R_1}\right) \left(\frac{R_4}{R_3}\right) = 1000$$



# Instrumentation Amplifier - Validation



$$R1 (R_{gain}) = R4 = R5 = 1K\Omega$$

$$R2 = R3 = 4.5 K\Omega$$

$$R6 = R7 = 100K\Omega$$

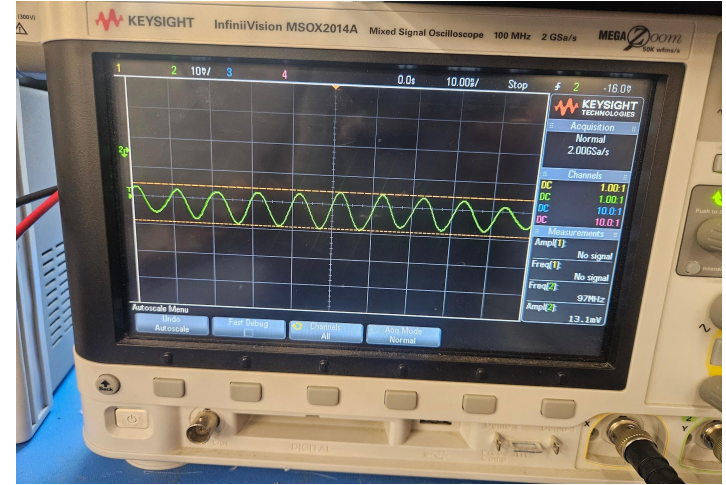
# Instrumentation Amplifier - Results

## Testing results:

- *Unsuccessful*
  - Input signal is not displayed on the digital oscilloscope.

## Challenges:

- Connection of waveform generator to the circuit (input signal)
  - Require: T-connector

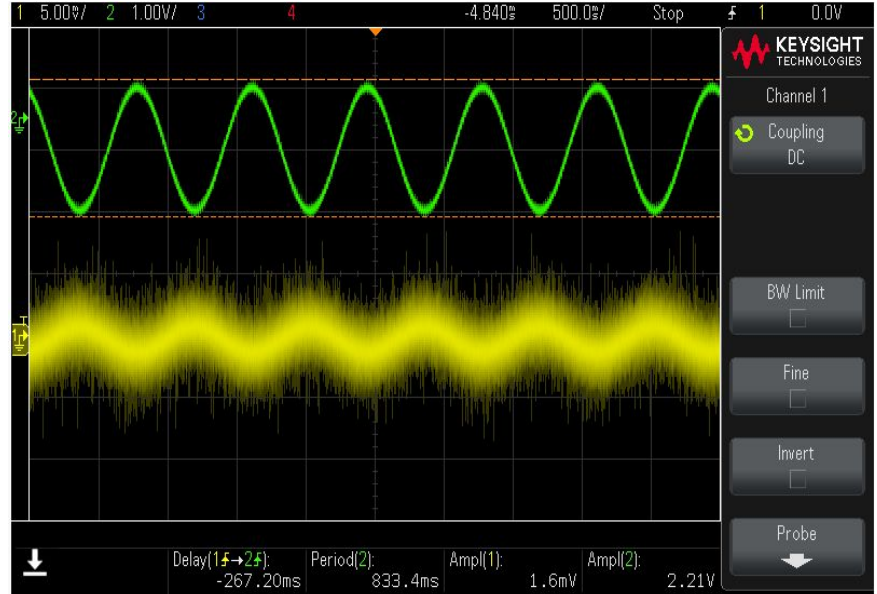




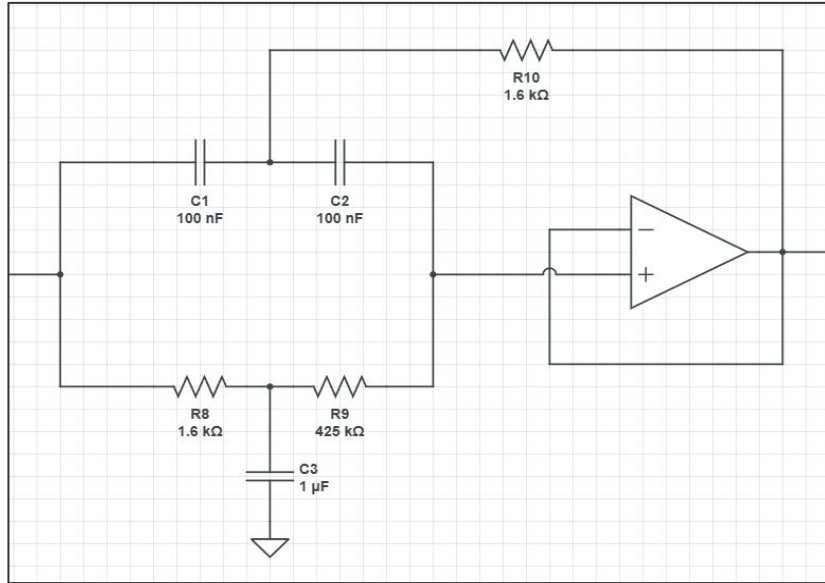
# Instrumentation Amplifier - Results

## Expected results:

- Yellow: Input Signal
  - Noisy signal
- Green: Output Signal
  - Amplified signal
  - Improved signal-to-noise ratio



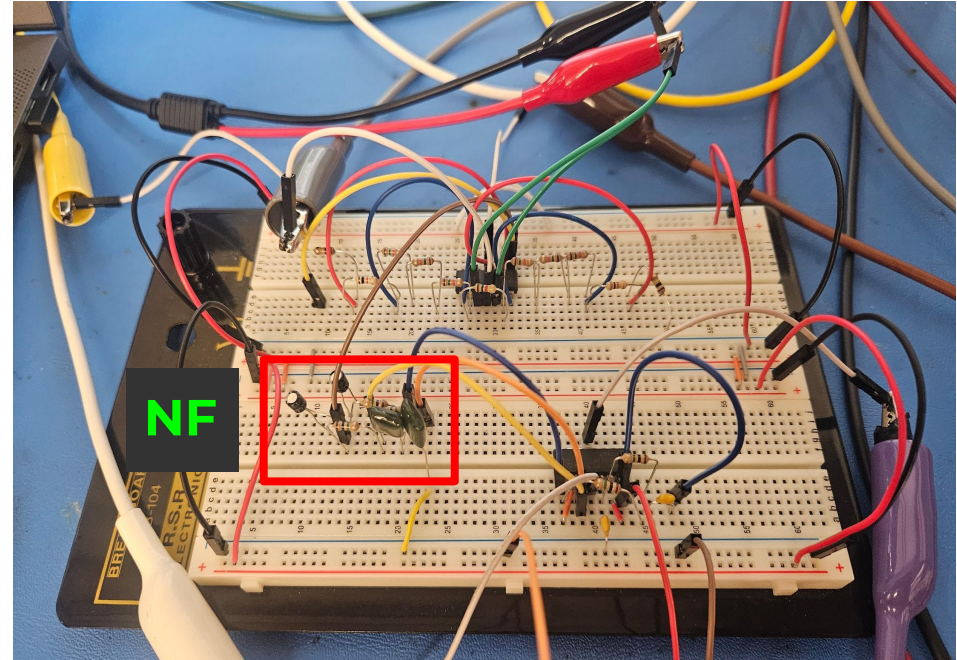
# Notch Filter Setup



$$R8 = R10 = 1.6K\Omega$$

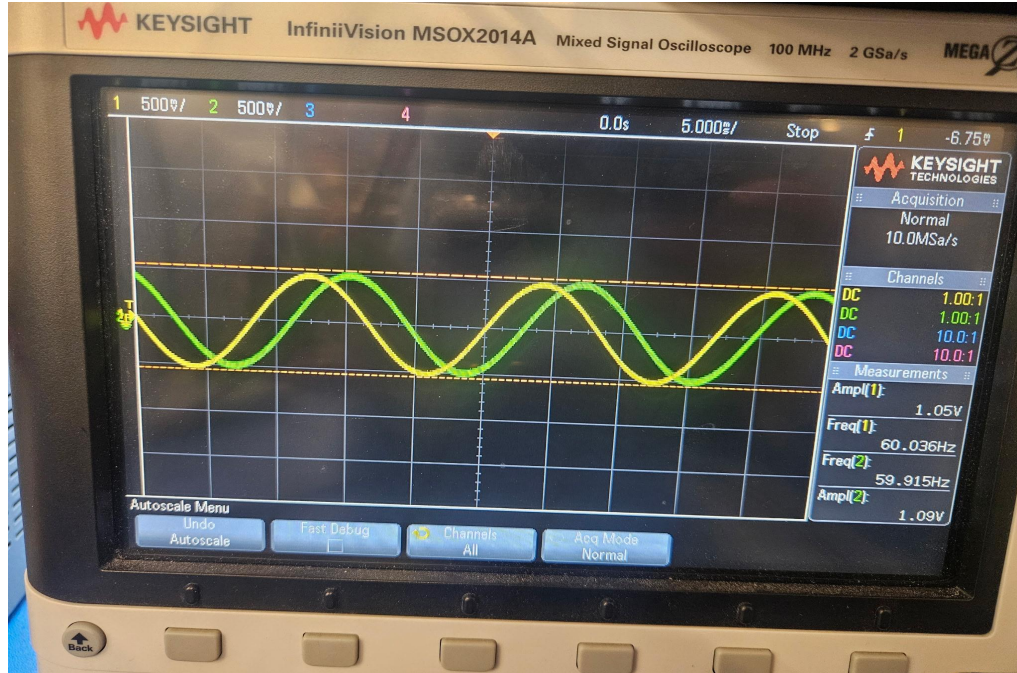
$$R9 = 425 K\Omega$$

$$C1 = C2 = 100nF ; C3 = 1\mu F$$



Cut-Off Frequency = 60 Hz

# Notch Filter - Validation



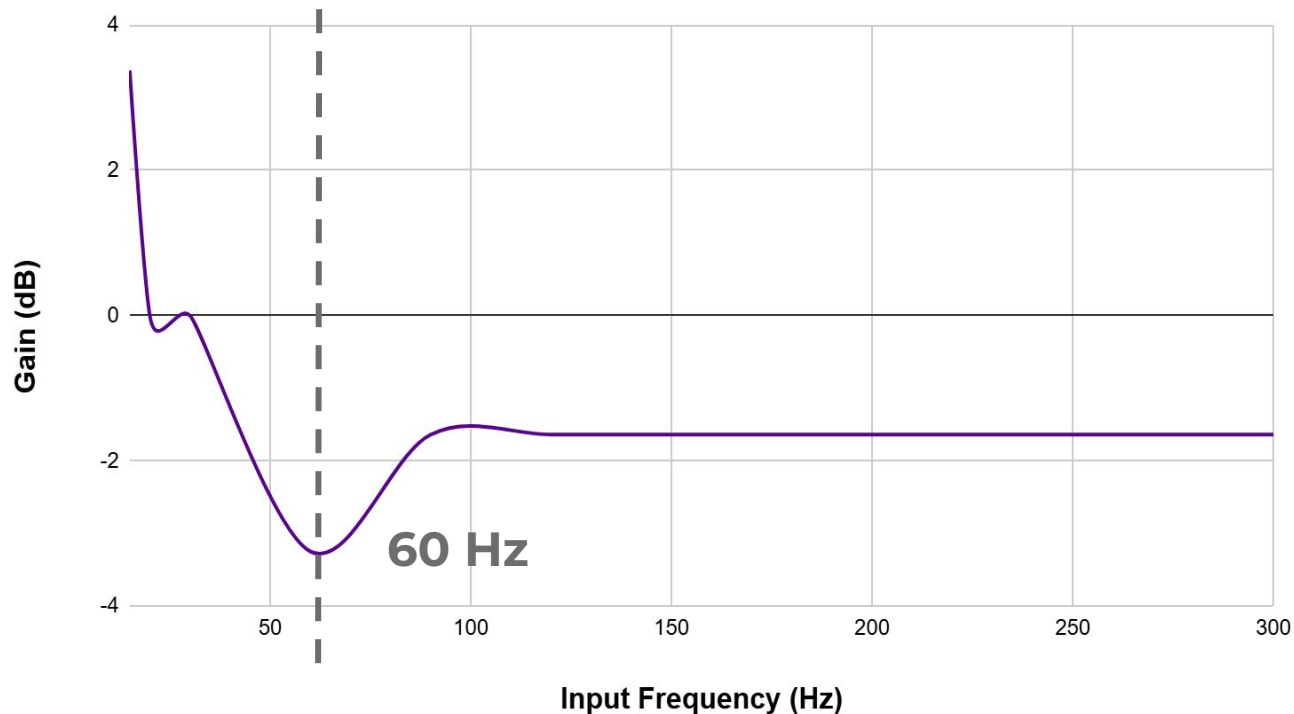
Green: Output Signal (1.09V)  
Yellow: Input Signal (1.05V)

Input frequency: 60.036 Hz  
Output frequency: 59.915 Hz

Attenuation observed

- Reducing 60 Hz interference

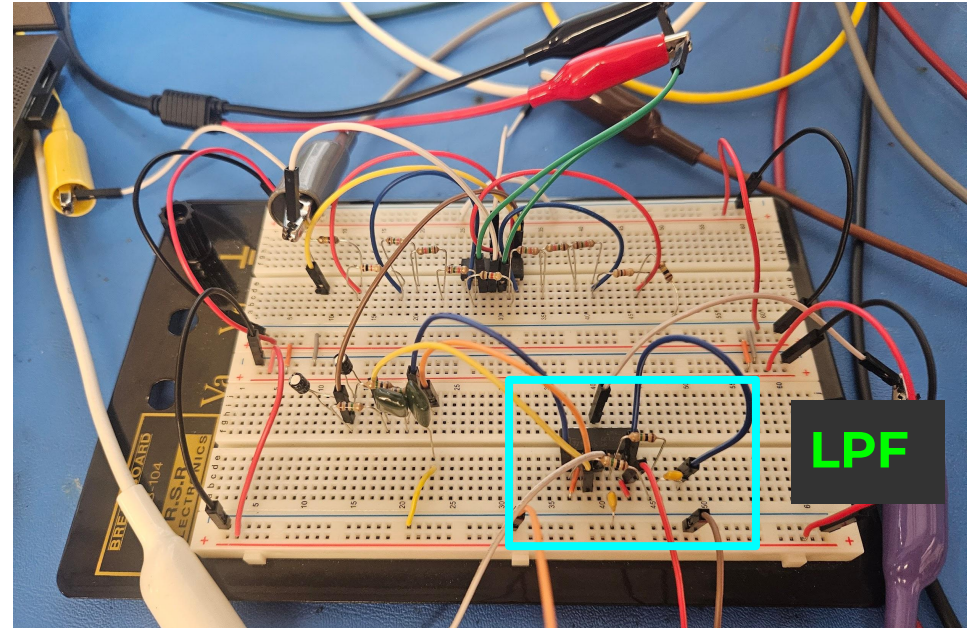
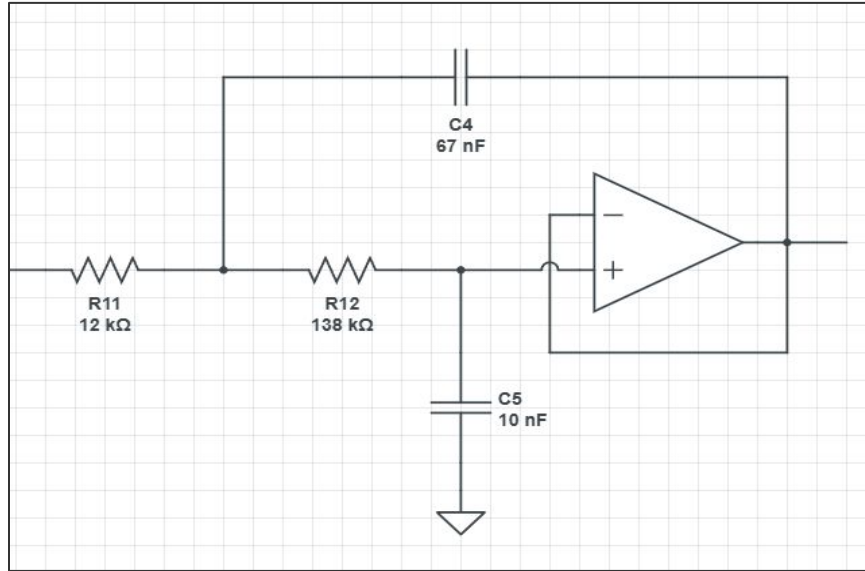
## Notch Filter Frequency Response Curve



- Targeted 60 Hz attenuation achieved
- Gain dips  $\sim -3.5$  dB at 60 Hz
- Minimal impact on other frequencies



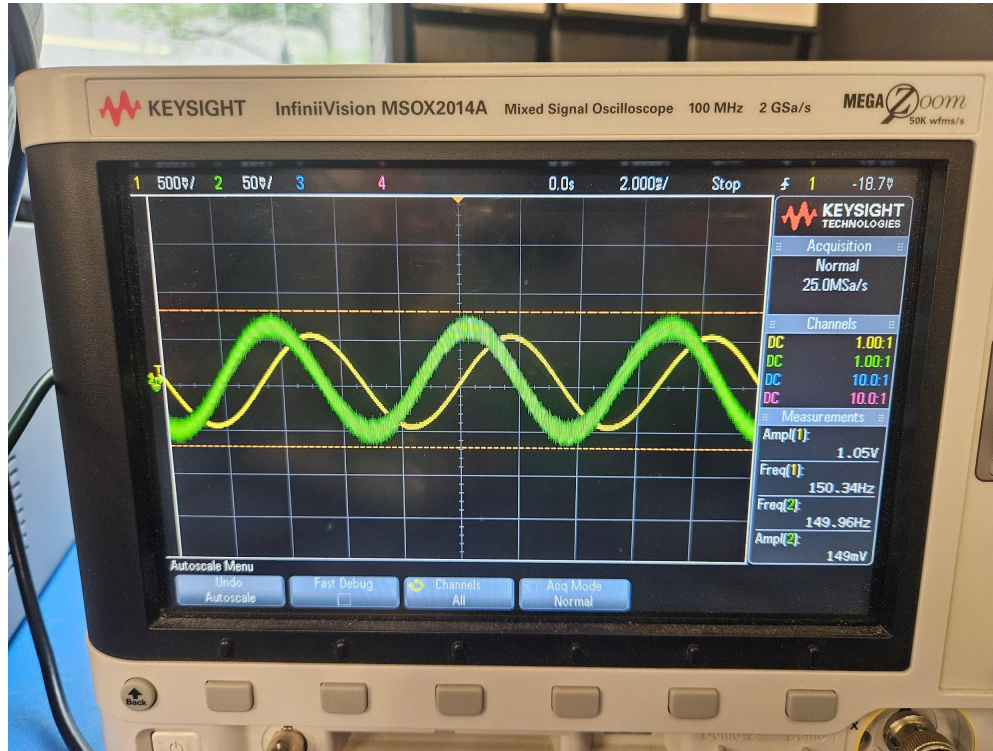
# Low Pass Filter Setup



$R11 = 12\text{K}\Omega$   
 $R12 = 138\text{K}\Omega$   
 $C4 = 67\text{nF}$  ;  $C5 = 10\text{nF}$

Cut-Off Frequency = 150 Hz

# Low Pass Filter Results



Green: Output Signal (149mV)

Yellow: Input Signal (1.05V)

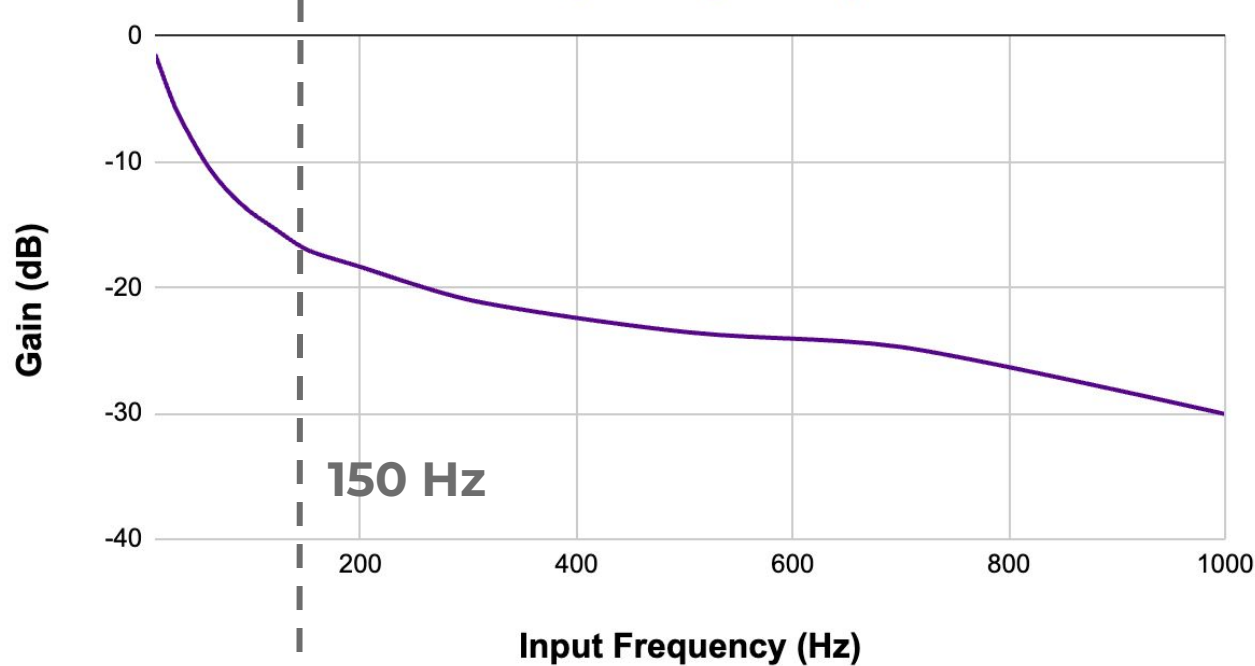
Input Frequency ~150 Hz

Output Frequency ~149.96 Hz

Attenuation of high frequency noise observed

- Amplitude reduced to just 149mV
- Suppression of out-of-band signals while preserving ECG-relevant components

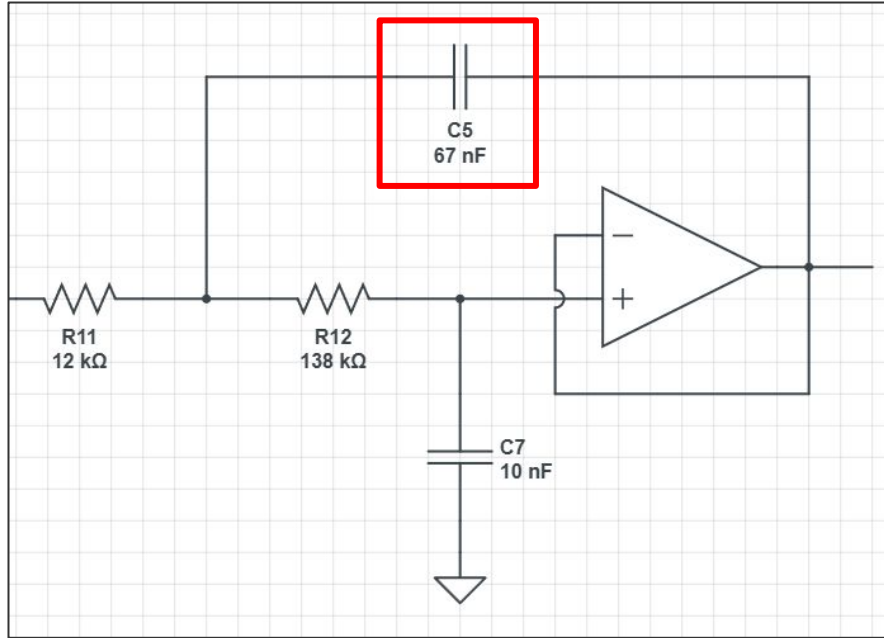
## Low Pass Filter Frequency Response Curve



- Cutoff frequency: 150 Hz
- Signals below 150 Hz pass
  - Expectation: flat gain (0 dB) below 150 Hz
- Signals above 150 Hz are attenuated



# Low Pass Filter Setup



## Challenges:

- Capacitor #5 replaced with 10nF

## Proposed solution and further testing:

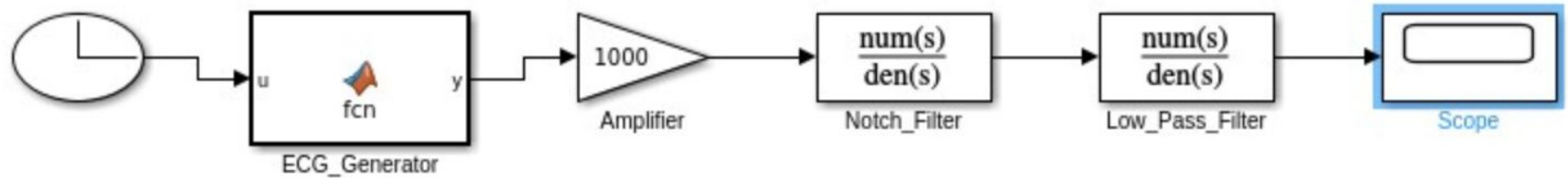
- Change Resistor #11 and #12 values to compensate for the replaced capacitor

# Software Verification

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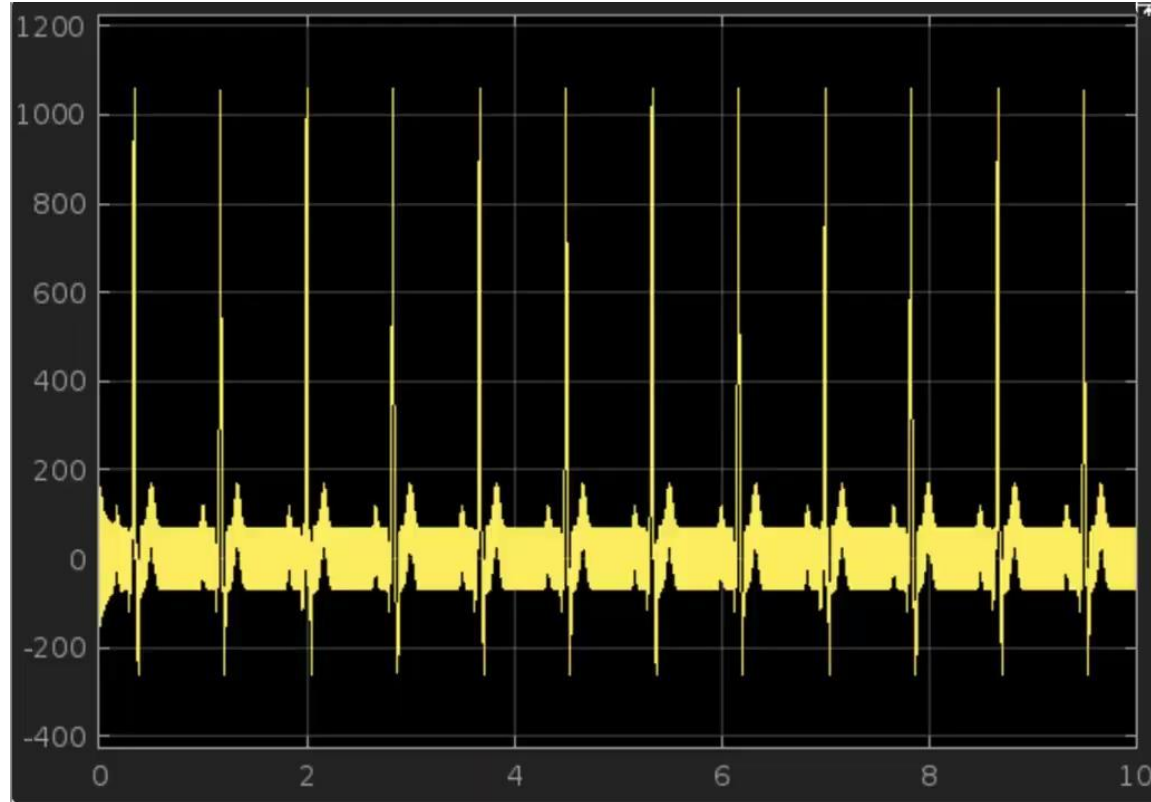
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# Simulink Model



| Component              | Simulation Function                    | What It Fixes                                              |
|------------------------|----------------------------------------|------------------------------------------------------------|
| Amplifier (x 1000)     | Boosts ECG signal to usable size       | Heart signals are too small (mV range)                     |
| 60 Hz Notch Filter     | Removes AC powerline noise at 60 Hz    | Electromagnetic noise contamination                        |
| 150 Hz Low-Pass Filter | Cuts high-frequency noise above 150 Hz | Muscle noise, artifacts, other high-frequency disturbances |

# Non-Ideal Testing - *Simulink*



# Components Integration

```
x = x + dx;  
add_block('simulink/Math Operations/Gain', [modelName '/Amplifier'], ...  
    'Position', [x y x+60 y+40], 'Gain', '1000');
```

$H_{\text{Notch}}$

```
x = x + dx;  
add_block('simulink/Continuous/Transfer Fcn', [modelName '/Notch_Filter'], ...  
    'Position', [x y x+80 y+40]);  
set_param([modelName '/Notch_Filter'], ...  
    'Numerator', '[1 0 142129]', ...  
    'Denominator', '[1 37.7 142129]');
```

$H_{\text{Low Pass}}$

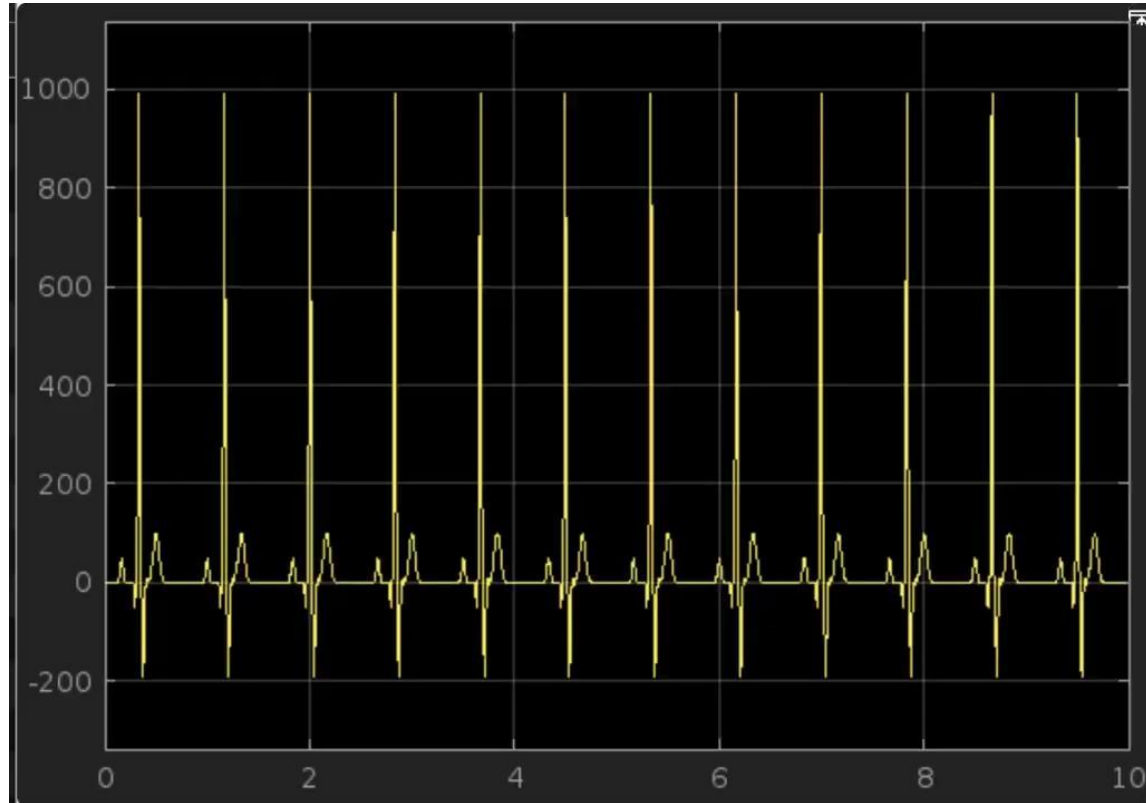
```
x = x + dx;  
add_block('simulink/Continuous/Transfer Fcn', [modelName '/Low_Pass_Filter'], ...  
    'Position', [x y x+80 y+40]);  
set_param([modelName '/Low_Pass_Filter'], ...  
    'Numerator', '[8896455]', ...  
    'Denominator', '[1 1884.96 8896455]');
```

# Ideal Testing - *Simulink*





# Ideal Testing - *Simulink*



# Conclusions

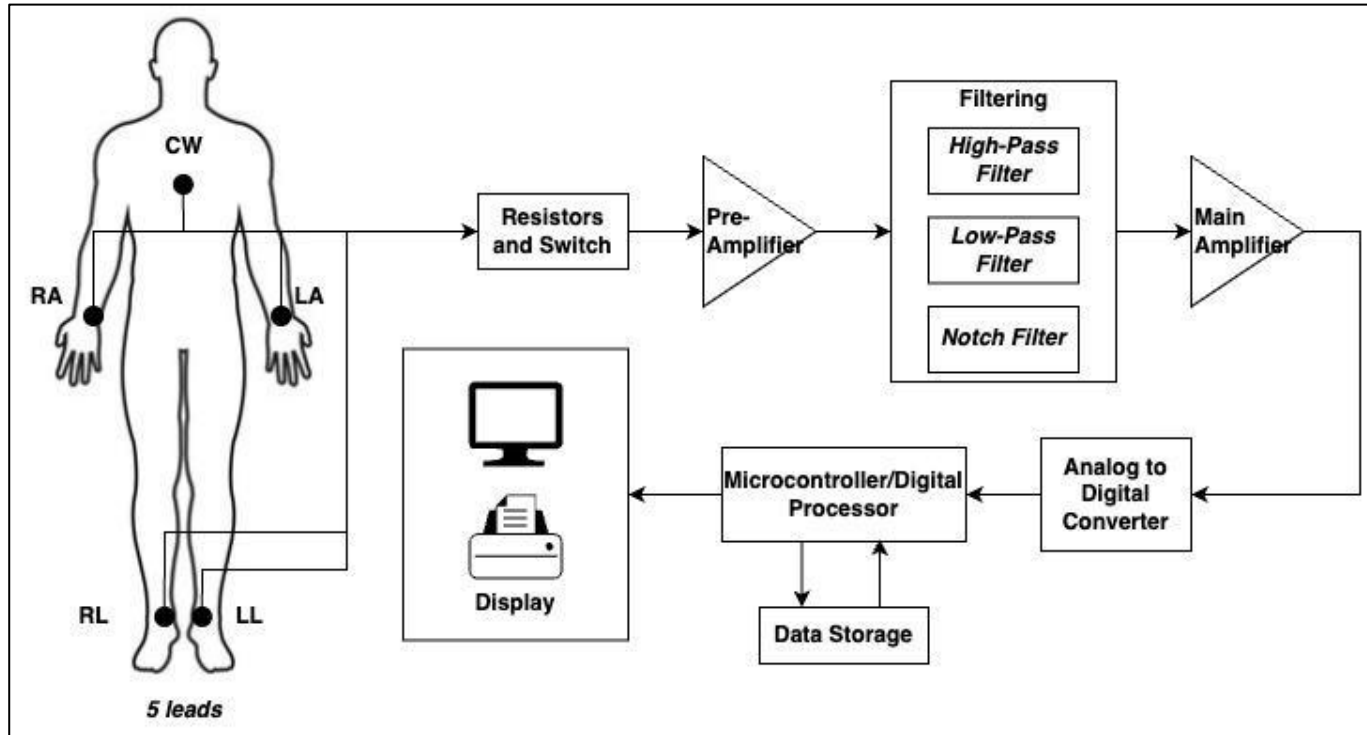
06.

# Conclusion

- **Confirmed** that early-stage signal conditioning minimizes the dependency on complex post-processing
  - Clinical needs for real-time analysis.
- **Established** a practical and scalable foundation for future development of more advanced ECG processing circuits.
  - Hardware-level enhancements to improve ECG signal acquisition quality
- **Validated** that amplification, notch filtering, and low-pass filtering can collectively address key noise and signal clarity challenges in resting ECG systems (Simulink).
  - Hardware requires improvements

**Thank you!**

# ECG System Components



# ECG System Circuit

## Notch Filter

$$R_1 = \frac{1}{2Q\omega_0 C} \quad R_2 = \frac{2Q}{\omega_0 C} \quad R_3 = \frac{R_1 R_2}{R_1 + R_2} \quad Q = \frac{\omega_0}{\beta} \quad \beta = \omega_{c2} - \omega_{c1}$$

## Low-Pass Filter

$$R_1 = \frac{2}{\omega_c [aC_2 + \sqrt{([a^2 + 4b(K-1)]C_2^2 - 4bC_1C_2)}]}$$

$$R_2 = \frac{1}{bC_1C_2R_1\omega_c^2}$$

$$C_1 \leq \frac{C_2[a^2 + 4b(K-1)]}{4b}$$

$$C_2 = \frac{10}{f_c} \mu F \text{ (preferably)}$$

$$a = 1.414214, b = 1, k = 1$$